

一、[参考解析] 设平面 P 上的抛物线 C 的顶点为 $X_0 = (x_0, y_0, z_0)$. 取平面 P 上 X_0 处相互正交的两单位向量 $\vec{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$ 和 $\vec{\beta} = (\beta_1, \beta_2, \beta_3)$, 使得 $\vec{\beta}$ 是抛物线 C 在平面 P 上的对称轴方向, 则抛物线的参数方程为

$$X(t) = X_0 + t\vec{\alpha} + \lambda t^2\vec{\beta}, \quad t \in \mathbb{R},$$

λ 为不等于 0 的常数.

记 $X(t) = (x(t), y(t), z(t))$, 则

$$x(t) = x_0 + \alpha_1 t + \lambda \beta_1 t^2, \quad y(t) = y_0 + \alpha_2 t + \lambda \beta_2 t^2, \quad z(t) = z_0 + \alpha_3 t + \lambda \beta_3 t^2.$$

因为 $X(t)$ 落在单叶双曲面 Γ 上, 代入方程 $x^2 + y^2 - z^2 = 1$, 得到对任意 t 要满足的方程

$$\lambda^2(\beta_1^2 + \beta_2^2 - \beta_3^2)t^4 + 2\lambda(\alpha_1\beta_1 + \alpha_2\beta_2 - \alpha_3\beta_3)t^3 + A_1t^2 + A_2t + A_3 = 0,$$

其中 A_1, A_2, A_3 是与 $X_0, \vec{\alpha}, \vec{\beta}$ 相关的常数. 于是得到

$$\beta_1^2 + \beta_2^2 - \beta_3^2 = 0, \quad \alpha_1\beta_1 + \alpha_2\beta_2 - \alpha_3\beta_3 = 0.$$

因为 $\vec{\alpha}, \vec{\beta}$ 是平面 P 上正交的单位向量, 所以有

$$\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 1, \quad \beta_1^2 + \beta_2^2 + \beta_3^2 = 1, \quad \alpha_1\beta_1 + \alpha_2\beta_2 + \alpha_3\beta_3 = 0.$$

于是得到

$$\beta_1^2 + \beta_2^2 = \beta_3^2 = \frac{1}{2}, \quad \alpha_1\beta_1 + \alpha_2\beta_2 = 0, \quad \alpha_3 = 0, \quad \alpha_1^2 + \alpha_2^2 = 1,$$

$$\vec{\alpha} = (\alpha_1, \alpha_2, 0), \quad \vec{\beta} = \left(-\frac{\varepsilon}{\sqrt{2}}\alpha_2, \frac{\varepsilon}{\sqrt{2}}\alpha_1, \beta_3\right), \quad \varepsilon = \pm 1.$$

于是得到平面 P 的法向量为 $\vec{n} = \vec{\alpha} \times \vec{\beta} = \left(A, B, \frac{\varepsilon}{\sqrt{2}}\right)$, 它与 z 轴方向 $\vec{e} = (0, 0, 1)$ 的夹角 θ 满足 $\cos \theta = \vec{n} \cdot \vec{e} = \pm \frac{1}{\sqrt{2}}$, 所以夹角为 $\frac{\pi}{4}$ 或 $\frac{3\pi}{4}$.

二、[参考解析] 充分性: 若 $\{a_n\}$ 有界, 则可设 $a_n \leq M$. 显然

$$\sum_{n=1}^m \frac{a_{n+1} - a_n}{a_n \ln a_{n+1}} \leq \sum_{n=1}^m \frac{a_{n+1} - a_n}{a_1 \ln a_1} = \frac{a_{m+1} - a_1}{a_1 \ln a_1} \leq \frac{M}{a_1 \ln a_1}.$$

由此可知 $\sum_{n=1}^{\infty} \frac{a_{n+1} - a_n}{a_n \ln a_{n+1}}$ 收敛.

必要性: 设 $\sum_{n=1}^{\infty} \frac{a_{n+1} - a_n}{a_n \ln a_{n+1}}$ 收敛. 由于

$$\ln a_{n+1} - \ln a_n = \ln \left(1 + \frac{a_{n+1} - a_n}{a_n} \right) \leq \frac{a_{n+1} - a_n}{a_n},$$

所以 $\frac{b_{n+1} - b_n}{b_{n+1}} \leq \frac{a_{n+1} - a_n}{a_n \ln a_{n+1}}$, 其中 $b_n = \ln a_n$. 因此, 级数 $\sum_{n=0}^{\infty} \frac{b_{n+1} - b_n}{b_{n+1}}$ 收敛.

由 Cauchy 收敛准则, 存在自然数 m , 使得对一切自然数 p , 有

$$\frac{1}{2} > \sum_{n=m}^{m+p} \frac{b_{n+1} - b_n}{b_{n+1}} \geq \sum_{n=m}^{m+p} \frac{b_{n+1} - b_n}{b_{m+p+1}} = \frac{b_{m+p+1} - b_m}{b_{m+p+1}} = 1 - \frac{b_m}{b_{m+p+1}}.$$

由此可知 $\{b_n\}$ 有界, 因为 p 是任意的, 所以 $\{a_n\}$ 有界.

题中级数的分母 a_n 不能换成 a_{n+1} . 例如: $a_n = e^{n^2}$ 无界, 但 $\sum_{n=1}^{\infty} \frac{a_{n+1} - a_n}{a_{n+1} \ln a_{n+1}}$ 收敛.

三、[参考证明] 必要性由迹的性质直接得.

充分性: 首先, 对于可逆矩阵 $W \in \Gamma$, 有 $WW_1, WW_2, \dots, WW_r$ 各不相同. 故有

$$W\Gamma \equiv \{WW_1, WW_2, \dots, WW_r\} = \{W_1, W_2, \dots, W_r\},$$

即 $W\Gamma = \Gamma, \forall W \in \Gamma$.

记 $S = \sum_{i=1}^r W_i$, 则 $WS = S, \forall W \in \Gamma$. 进而 $S^2 = rS$, 即 $S^2 - rS = 0$. 若 λ 为 S 的特征值, 则 $\lambda^2 - r\lambda = 0$, 即 $\lambda = 0$ 或 $\lambda = r$.

结合条件 $\sum_{i=1}^r \text{tr}(W_i) = 0$ 知, S 的特征值只能为 0. 因此 $S - rI$ 可逆 (例如取 S 的 Jordan 分解就可以直接看出).

再次注意到 $S(S - rI) = S^2 - rS = O$, 此时右乘 $(S - rI)^{-1}$, 即得 $S = O$.

四、[参考证明] 反证法: 若 $XN + Y^T M^T = O$, 则 $N^T X^T + MY = O$. 另外, 由 $(X, Y) \in T$ 得 $XY + (XT)^T = 2aI$, 即 $XY + Y^T X^T = 2aI$. 类似有 $MN + N^T M^T = 2aI$. 因此

$$\begin{pmatrix} X & Y^T \\ M & N^T \end{pmatrix} \begin{pmatrix} Y & N \\ X^T & M^T \end{pmatrix} = 2a \begin{pmatrix} I & O \\ O & I \end{pmatrix}.$$

进而 $\frac{1}{2a} \begin{pmatrix} Y & N \\ X^T & M^T \end{pmatrix} \begin{pmatrix} X & Y^T \\ M & N^T \end{pmatrix} = \begin{pmatrix} I & O \\ O & I \end{pmatrix}$, 得 $YY^T + NN^T = O$, 所以 $Y = O, N = O$. 导致 $XY = O$, 与 $XY = aI + A \neq O$ 矛盾.

五、[参考解析][思路一] 由定积分的定义, 有

$$\begin{aligned} A &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) = \int_0^1 f(x) \, dx \\ &= x \arctan x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx = \frac{\pi}{4} - \frac{\ln 2}{2}. \end{aligned}$$

对于 $x \in \left(\frac{k-1}{n}, \frac{k}{n}\right)$ ($1 \leq k \leq n$), 由中值定理, 存在 $\xi_{n,k} \in \left(\frac{k-1}{n}, \frac{k}{n}\right)$ 使得

$$f(x) = f\left(\frac{k}{n}\right) + f'\left(\frac{k}{n}\right)\left(x - \frac{k}{n}\right) + \frac{f''(\xi_{n,k})}{2}\left(x - \frac{k}{n}\right)^2.$$

于是

$$\begin{aligned} &\left| \sum_{k=1}^n f\left(\frac{k}{n}\right) - nA + \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f'\left(\frac{k}{n}\right)\left(x - \frac{k}{n}\right) \, dx \right| \\ &= \left| \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left[f\left(\frac{k}{n}\right) + f'\left(\frac{k}{n}\right)\left(x - \frac{k}{n}\right) - f(x) \right] \, dx \right| \\ &\leq M \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left(x - \frac{k}{n}\right)^2 \, dx = \frac{M}{3n}, \end{aligned}$$

其中 $M = \frac{1}{2} \max_{x \in [0,1]} |f''(x)|$. 因此

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n f\left(\frac{k}{n}\right) - An \right) &= - \lim_{n \rightarrow \infty} \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f'\left(\frac{k}{n}\right)\left(x - \frac{k}{n}\right) \, dx \\ &= \lim_{n \rightarrow \infty} \frac{1}{2n} \sum_{k=1}^n f'\left(\frac{k}{n}\right) = \frac{1}{2} \int_0^1 f'(x) \, dx = \frac{\pi}{8}. \end{aligned}$$

[思路二] 由定积分的定义, 有

$$\begin{aligned} A &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) = \int_0^1 f(x) \, dx \\ &= x \arctan x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx = \frac{\pi}{4} - \frac{\ln 2}{2}. \end{aligned}$$

对于 $x \in \left(\frac{k-1}{n}, \frac{k}{n}\right)$ ($1 \leq k \leq n$), 由中值定理, 存在 $\xi_{n,k} \in \left(\frac{k-1}{n}, \frac{k}{n}\right)$ 使得

$$f\left(\frac{k}{n}\right) = f(x) + f'(x)\left(\frac{k}{n} - x\right) + \frac{f''(\xi_{n,k})}{2}\left(\frac{k}{n} - x\right)^2.$$

于是

$$\begin{aligned} & \left| \sum_{k=1}^n f\left(\frac{k}{n}\right) - nA - \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f'(x) \left(\frac{k}{n} - x\right) dx \right| \\ &= \left| \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left[f\left(\frac{k}{n}\right) - f(x) - f'(x) \left(\frac{k}{n} - x\right) \right] dx \right| \\ &\leq M \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left(\frac{k}{n} - x\right)^2 dx = \frac{M}{3n}, \end{aligned}$$

其中 $M = \frac{1}{2} \max_{x \in [0,1]} |f''(x)|$. 因此

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n f\left(\frac{k}{n}\right) - An \right) \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f'(x) \left(\frac{k}{n} - x\right) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n n f'(\eta_{n,k}) \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left(\frac{k}{n} - x\right) dx \\ &= \lim_{n \rightarrow \infty} \frac{1}{2n} \sum_{k=1}^n f'(\eta_{n,k}) = \frac{1}{2} \int_0^1 f'(x) dx = \frac{\pi}{8}, \end{aligned}$$

其中 $\eta_{n,k} \in \left(\frac{k-1}{n}, \frac{k}{n}\right)$.

[思路三] 由定积分定义, 有

$$\begin{aligned} A &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) = \int_0^1 f(x) dx \\ &= x \arctan x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} dx = \frac{\pi}{4} - \frac{\ln 2}{2}. \end{aligned}$$

对于 $x \in \left(\frac{k-1}{n}, \frac{k}{n}\right)$ ($1 \leq k \leq n$), 由中值定理, 存在

$$\xi_{n,k} \in \left(\frac{k-1}{n}, \frac{k}{n}\right)$$

使得

$$f(x) = f\left(\frac{k}{n}\right) + f'\left(\frac{k}{n}\right) \left(x - \frac{k}{n}\right) + \frac{f''(\xi_{n,k})}{2} \left(x - \frac{k}{n}\right)^2.$$

于是

$$\begin{aligned}
 & \left| \sum_{k=1}^n f\left(\frac{k}{n}\right) - nA - n \int_1^{1+\frac{1}{2n}} f(x) \, dx + n \int_0^{\frac{1}{2n}} f(x) \, dx \right| \\
 &= \left| \sum_{k=1}^n n \int_{\frac{k-\frac{1}{2}}{n}}^{\frac{k+\frac{1}{2}}{n}} \left[f\left(\frac{k}{n}\right) - f(x) + f'\left(\frac{k}{n}\right) \left(\frac{k}{n} - x\right) \right] \, dx \right| \\
 &\leq M \sum_{k=1}^n n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left(\frac{k}{n} - x\right)^2 \, dx = \frac{M}{3n},
 \end{aligned}$$

其中 $M = \frac{1}{2} \max_{x \in [0,1]} |f''(x)|$. 因此

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n f\left(\frac{k}{n}\right) - An \right) &= \lim_{n \rightarrow \infty} n \int_1^{1+\frac{1}{2n}} f(x) \, dx - \lim_{n \rightarrow \infty} n \int_0^{\frac{1}{2n}} f(x) \, dx \\
 &= \frac{f(1)}{2} - \frac{f(0)}{2} = \frac{\pi}{8}.
 \end{aligned}$$

六、[参考解析] 容易知道 $f(x)$ 连续. 注意到 $f(x) = 1 - x^2(1-x)$, 于是有

$$0 < f(x) < 1 = f(0) = f(1), \quad \forall x \in (0, 1).$$

任取 $\delta \in \left(0, \frac{1}{2}\right)$, 有 $\eta = \eta_\delta \in (0, \delta)$ 使得

$$m_\eta = \min_{x \in [0, \eta]} f(x) > M_\delta \equiv \max_{x \in [\delta, 1-\delta]} f(x).$$

于是当 $n \geq \frac{1}{\delta^2}$ 时,

$$\begin{aligned}
 0 &\leq \frac{\int_\delta^1 f^n(x) \, dx}{\int_0^\delta f^n(x) \, dx} = \frac{\int_{1-\delta}^1 f^n(x) \, dx}{\int_0^\delta f^n(x) \, dx} + \frac{\int_\delta^{1-\delta} f^n(x) \, dx}{\int_0^\delta f^n(x) \, dx} \\
 &= \frac{\int_0^\delta (1-x(1-x)^2)^n \, dx}{\int_0^\delta (1-x^2(1-x))^n \, dx} + \frac{\int_\delta^{1-\delta} f^n(x) \, dx}{\int_0^\delta f^n(x) \, dx} \\
 &\leq \frac{\int_0^\delta \left(1 - \frac{x}{4}\right)^n \, dx}{\int_0^\delta (1-x^2)^n \, dx} + \frac{\int_\delta^{1-\delta} f^n(x) \, dx}{\int_0^\eta f(x) \, dx} \leq \frac{\int_0^\delta \left(1 - \frac{x}{4}\right)^n \, dx}{\int_0^{\frac{1}{\sqrt{n}}} \left(1 - \frac{x}{\sqrt{n}}\right)^n \, dx} + \frac{(1-2\delta)M_\delta^n}{\eta m_\eta^n}
 \end{aligned}$$

$$= \frac{\frac{4}{n+1} \left(1 - \left(1 - \frac{\delta}{4} \right)^{n+1} \right)}{\frac{\sqrt{n}}{n+1} \left(1 - \left(1 - \frac{1}{n} \right)^{n+1} \right)} + \frac{(1-\delta)}{\eta} \left(\frac{M_\delta}{m_\eta^n} \right)^n.$$

从而 $\lim_{n \rightarrow \infty} \frac{\int_\delta^1 f^n(x) \, dx}{\int_0^\delta f^n(x) \, dx} = 0.$

对于 $\varepsilon \in \left(0, \ln \frac{5}{4} \right)$, 取 $\delta = 2(e^\varepsilon - 1)$, 则 $\delta \in \left(0, \frac{1}{2} \right)$, $\ln \frac{2+\delta}{2} = \varepsilon.$

另一方面, 由前述结论, 存在 $N \geq 1$ 使得当 $n \geq N$ 时, 有 $\frac{\int_\delta^1 f^n(x) \, dx}{\int_0^\delta f^n(x) \, dx} \leq \varepsilon.$ 从

而又有

$$\begin{aligned} & \left| \frac{\int_0^1 f^n(x) \ln(x+2) \, dx}{\int_0^1 f^n(x) \, dx} - \ln 2 \right| \\ &= \frac{\int_0^1 f^n(x) \ln \frac{x+2}{2} \, dx}{\int_0^1 f^n(x) \, dx} \\ &\leq \frac{\int_0^\delta f^n(x) \ln \frac{x+2}{2} \, dx}{\int_0^\delta f^n(x) \, dx} + \frac{\int_\delta^1 f^n(x) \ln \frac{x+2}{2} \, dx}{\int_0^\delta f^n(x) \, dx} \\ &\leq \ln \frac{\delta+2}{2} + \frac{\ln 2 \int_\delta^1 f^n(x) \, dx}{\int_0^\delta f^n(x) \, dx} \leq \varepsilon(1 + \ln 2). \end{aligned}$$

因此, $\lim_{n \rightarrow \infty} \frac{\int_0^1 f^n(x) \ln(x+2) \, dx}{\int_0^1 f^n(x) \, dx} = \ln 2.$